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# Evolution - 100 MCQs

NCERT Nichod : Biology Practice Series NEET 2025

**Anmol Sharma** • Lesson 23 • Feb 3, 2025

10AM - Class 11th

Good afternoon
↓
audible?
↓
Ready?

# Evolution

BIOLOGYLIVE

# Title: Evolution

1. Evolutionary history of an organism is also known as

(1) Phylogeny

(2) Paleontology ⇒ Fossil Study

(3) Ontogeny

(4) Phycology

# Title: Evolution

**2. According to Lamarckism, long necked Giraffes evolved because**

(1) Humans preferred only long necked ones

(2) Short necks suddenly changed into long necks

(3) Nature selected only long necked ones

(4) Stretching of necks over many generation by short necked ones

# Title: Evolution

V.Imp

3. In a hypothetical human population in HardyWeinberg equilibrium, the frequencies of the alleles IA, IB and i are respectively 0.1, 0.3 and 0.6 . What percentage of the population is expected to have 'B' type blood?

(1) 27

(2) 9

(3) 36

(4) 45

V.Imp

$I^A = 0.1$

$I^B = 0.3$

$i = 0.6$

Blood Group B =

$= I^B I^B + I^B i$

$= (0.3 \times 0.3) + (0.3 \times 0.6)$

$= 0.09 + 0.18$

$= 0.27$

$= 27\%$

# Title: Evolution

**4. Which of the following are the two key concepts of Darwinian theory of evolution?**

(1) Branching descent and mutation

(2) Adaptive radiation and branching ascent

(3) Mutation and natural selection

(4) Branching descent and natural selection

v.Imp

5. Given below are some stages of human evolution. Arrange them in correct sequence. "(Past to Recent)"

A. Homo habilis 1

B. Homo sapiens 4

C. Homo neanderthalensis 3

D. Homo erectus 2

Choose the correct sequence of human evolution from the options given below:

(1) D - A - C - B

(2) B - A - D - C

(3) C - B - D - A

(4) A - D - C - B

**6. Evolution of man took place in :-**

(1) Australia

(2) Africa

(3) Central America

(4) Central Asia

**7. If the gene flow occurs by chance, it is called?**

(1) Genetic shift

(2) Genetic drift

(3) Genetic migration

(4) None of them

**8. In which respect does Darwin's theory is wrong?**

(1) Survival of the fittest

survival

(2) ~~Arriv~~al of the fittest

(3) High efficiency of reproduction

(4) Origin of species

# Title: Evolution



9. "Given below are four statements (A - D) each with one or two blanks. Select the option which correctly fills up the blanks in two statements.

(A) Wings of butterfly and birds look alike and are the results of ____(i)____volution.

convergent

(B) Miller showed that $CH_4$, $H_2$, $NH_3$ and ____(i)____ when exposed to electric discharge in a flask resulted in formation of ____ (ii)____

W.V

A.C

(C) Vermiform appendix is a ____(i)____organ and an ____(ii)____evidence of evolution.

Ves

anat

(D) According to Darwin evolution took place due to ____(i)____and ____(ii)____ of the fittest.

Small variation Survival

Options:

(1) (A) - (i) convergent,
(B) - (i) Oxygen, (ii) nucleosides.

(2) (B) - (i) Water vapour, (ii) Amino acids,
(C) - (i) vestigial, (ii) Anatomical.

(3) (C) - (i) vestigial, (ii) anatomical,
(D) - (i) Mutations, (ii) Multiplication.

(4) (D) - (i) small variations, (ii) survival,
(C) - (i) convergent, (ii) adaptive

# Title: Evolution

**10. Which is amongst the following show common ancestry line of mammals?**

(1) Therapsids → 3 → mammals

(2) Synapsids → 1

(3) Pelycosaurs → 2

(4) All of these

# Title: Evolution

V.Imp

**11. Fill in the blanks :**

**Mutations are______ while Darwinian variations are______**

(1) Constant and directionless, small and directional.

(2) Random and directional, small and directional.

(3) Random and directionless, small and directional.

(4) Random and directionless, small and directionless

# Title: Evolution



**12. Who proved the theory of chemical evolution through his experiment?**

(1) L.S. Miller, 1953

(2) S.L. Miller, 1952

(3) S.L. Miller, 1953

(4) L.S. Miller, 1952

# Title: Evolution

**13. Statement A : Ernst Haeckel supported the embryological similarities as an evidence of evolution.**

**Statement B : The embryos of all vertebrates including human develop a row of vestigial gill slits which are absent in adult vertebrates.**

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

(3) Both statements are correct.

(4) Both Statements are incorrect.

V.Imp

**14. Which of the following statement is false?**

(1) Microbial experiments show that pre- existing advantageous mutations when selected will result in observation of new phenotypes. Over few generations, this would result in Speciation.

Single species

(2) The sum total of genes of ~~all the species~~ in an area constitutes the gene pool. X

(3) The gene frequency refers to the proportion of an allele in the gene pool as compared with other alleles.

(4) None of the above

# Title: Evolution

**15. The theory of natural selection given by Charles Darwin was further supported by**

(1) Lamarck

(2) Alfred Wallace

(3) Darwin

(4) Oparian and Haldane

**16. Random genetic drift in a population probably results from :-**

(1) Large population size

(2) Highly genetically variable individuals

(3) Constant low mutation rate

(4) Interbreeding within this population

V.Imp

**17. Among the following sets of examples for divergent evolution, select the incorrect option:**

(1) Brain of bat, man and cheetah

(2) Heart of bat, man and cheetah

(3) Forelimbs of man, bat and cheetah

(4) Eye of octopus, bat and man → analogous organ → convergent

**18. Which of the following is used as an atmospheric pollution indicator**

(1) Lepidoptera

(2) Lichens

(3) Lycopersicon

(4) Lycopodium

# Title: Evolution



19. In a population of plants, some ~~water~~ plant extremely tall and the remaining were extremely dwarf. No plants of the population showed intermediate height. The type of operation of natural selection in the above case is

(1) Balancing

(2) Directional

(3) Stabilizing

(4) Disruptive

**20. A potential damage to a population that has been greatly reduced in number is the:-**

(1) Loss of genetic variability (Bottle neck effect)

(2) Tendency towards assortative mating

(3) Reduced gene flow

(4) Hardy - Weinberg disequilibrium

# Title: Evolution



**21. The gene pool represents_____ in a population and it remains a constant.**

(1) Total genes only

(2) Dominant alleles only

(3) Recessive alleles only

(4) Total genes and their alleles

# Title: Evolution



v.Imp

**22. The change of the "lighter" coloured variety of peppered moth, Biston betularia to its darker variety (carbonaria) is due to**

mutation

(1) Deletion of a segment of genes due to industrial pollution

(2) Translocation of a block of genes in chromosomes in response to heavy carbons

(3) Industrial carbon deposited on the barks of the trees resulting in darker variety being more camouflaged

(4) Mutation of single mendelian gene for survival in smoke laden industrial environment.

# Title: Evolution

23. **Statement A : The skull of ~~adult~~ baby chimpanzee is more like adult human skull than ~~baby~~ adult chimpanzee skull.** ×

**Statement B : The skull of baby chimpanzee is more like adult human than adult chimpanzee skull.** ✓

(1) Statement A is correct but Statement B is incorrect.

✓ (2) Statement A is incorrect but Statement B is correct.

(3) Both statements are correct.

(4) Both Statements are incorrect.

**24. Statement A : Giant ferns fell to form coal deposits slowly.**

**Statement B : About 65 million years ago dinosaurs died out.**

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

(3) Both statements are correct.

(4) Both Statements are incorrect.

# Title: Evolution

→ Divergent evolution.

**25. When more than one adaptive radiation appeared to have occurred in an isolated geographical area, it is**

(1) Convergent evolution

(2) Divergent evolution

(3) Natural selection

(4) Genetic drift

# Title: Evolution

**26. Which of the following structures are 'homologous' from the evolutionary point of view?**

(1) Petals of rose and the anther of an apple blossom.

(2) Thorn of bougainvillea and tendrils of cucurbita

(3) Spines of a cactus and spines of a porcupine.

(4) Flagellum of Euglena and the flagella of Paramecium

**27. Dinosaurs became extinct during ____ time period.**

(1) Jurassic period

(2) Triassic period

(3) Permian Period

(4) Cretaceous period

# Title: Evolution

**28. The eye of octopus and eye of cat show different patterns of structure, yet they perform similar function. This is an example of**

(1) Homologous organs that have evolved due to divergent evolution.

(2) Analogous organs that have evolved due to convergent evolution.

(3) Analogous organs that have evolved due to divergent evolution.

(4) Homologous organs that have evolved due to convergent evolution.

**29. On critical analysis which of the following has resulted in changed frequency of genes and alleles in future generation?**

(1) Variation due to mutation

(2) Variation due to recombination

(3) Gene flow or genetic drift

(4) All of these

# Title: Evolution



**30. Who said that evolution of the life forms has occurred but driven by use and disuse of organs:**

(1) Darwin - French naturalist

(2) Lamarck - French naturalist

(3) De Vries - French naturalist

(4) Malthus - French naturalist

# Title: Evolution

**31. In a population that is in Hardy-Weinberg equilibrium, the frequency of the recessive homozygote genotype of a certain trait is 0.09 . Percentage of individuals homozygous for the dominant allele is**

(1) 0.09

(2) 0.30

(3) 0.70

(4) 0.49

# Title: Evolution

**32. Tasmanian Wolf is a marsupial while Wolf is a placental mammal. This is an example of**

(1) Convergent evolution

(2) Divergent evolution

(3) Inheritance of acquired characters

(4) Adaptive radiation

# Title: Evolution

**33. The process by which different type of finches were evolved in Galapagos islands is a result of**

(1) Adaptive radiation

(2) Non -adaptive radiation

(3) Industrial melanism

(4) Convergent evolution

# Title: Evolution

**34. Hardy-Weinberg law is applicable to**

(1) Small population

(2) Genetic equilibrium

(3) Genetic drift

(4) Non-random mating

# Title: Evolution

**35. S.L. Miller created electric discharge in a closed flask containing.**

(1) $CH_4$, $H_2$, $NH_3$ and water vapour at $80^0C$

(2) $CH_4$, $H_2$, $NH_3$ and water vapour at $800^0C$

(3) $CH_4$, $H_2$, $NH_3$ and water vapour at $8000^0C$

(4) $CH_4$, $H_2$, $NH_2$ and water vapour at $800^0C$

# Title: Evolution

**36. Variations during mutations of meiotic recombinations are**

(1) Random and directionless

(2) Random and directional

(3) Small and directional

(4) Random, small and directional

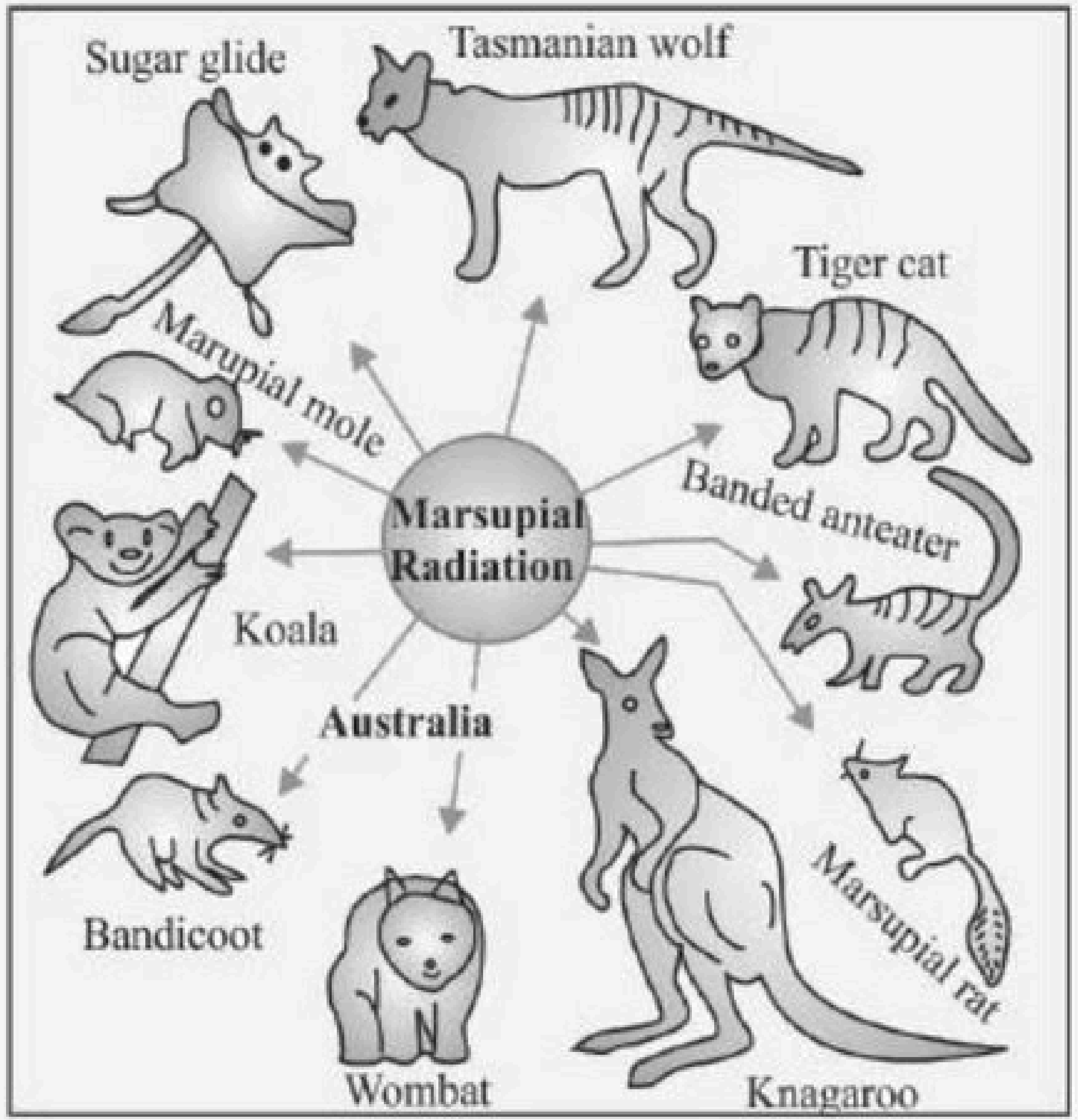
**37. The given figure represents**

(1) Adaptive Radiation of marsupials of Australia

(2) Convergent evolution of Homo erectus

(3) Divergent evolution of Marupials of Australia

(4) Family tree of Homo erectus



**38. Divergent evolution gives rise to**

(1) Homologous organs

(2) Analogous organs

(3) Both (1) and (2)

(4) None of these

# Title: Evolution



**39. Statement A : Sudden inheritable changes take place in genome of an organism due to certain factors called mutations.**

**Statement B : Lamarck's theory of evolution is popularly called the theory of continuity of germplasm.**

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

(3) Both statements are correct.

(4) Both Statements are incorrect.

**40. Example of convergent evolution is**

(1) Darwin finches and marsupial mouse

(2) Placental wolf and Tasmanian wolf

(3) Lemur and Spotted cuscus

(4) Both 2 and 3

**41. Millers experiment provided evidence for theory of**

(1) Biogenesis

(2) Abiogenesis

(3) Special creation

(4) Organic evolution

# Title: Evolution

**42. Analogous structures are a result of**

(1) Convergent evolution

(2) Shared ancestry

(3) Stabilizing selection

(4) Divergent evolution

# Title: Evolution

## 43. Match the following columns.

| Column I | | Column II | |
|---|---|---|---|
| A | Stabilisation | P | Industrial melanism |
| B | Directional selection | Q | More individuals possess heterozygous character |
| C | Disruption selection | R | More individuals possess homozygous character |

(1) A – R, B – P, C – Q

(2) A – Q, B – P, C – R

(3) A – P, B – R, C – Q

(4) A – Q, B – R, C – P

# Title: Evolution

**44. Match the column I with Column II**

| Column I | Column II |
| --- | --- |
| A Mesozoic | P First amphibians |
| B Devonian | Q 300-250 mya |
| C Palacocene | R 160 million years |
| D Permian | S Radiation of primitive mammals |

(1) A – R,B – P,C – R,D – Q

(2) A – R,B – P,C – S,D – Q

(3) A – P,B – R,C – Q,D – S

(4) A – Q,B – P,C – R,D – S

# Title: Evolution

## 45. Match the following columns.

| Column I | | Column II | |
|---|---|---|---|
| A | Genetic drift | P | Change in the population's allele frequency due to chance alone |
| B | Natural selection | Q | Selection of better traits |
| C | Gene flow | R | Immigation or emigration changes the allele frequency |
| D | Mutation | S | Saltation |

(1) A – R, B – P, C – Q, D – S
(2) A – S, B – R, C – P, D – Q
(3) A – P, B – Q, C – R, D – S
(4) A – Q, B – P, C – R, D – S

# Title: Evolution

**46. Match the scientists listed under column ' I ' with ideas listed Column 'II'**

| Column I | | Column II | |
|---|---|---|---|
| A | Darwin | P | Abiogenesis |
| B | Oparin | Q | Use and disuse of organs |
| C | Lamarck | R | Germ plasm theory and neodarwinism |
| D | August Weisman | S | Evolution by natural selection |

(1) A - P, B - S, C - Q, D – R

(2) A - S, B - P, C - Q, D - R

(3) A - Q, B - S, C - R, D – P

(4) A - S, B - R, C - Q, D - P

# Title: Evolution

**47. Match Column I with Column II:**

**Choose the correct answer from the options given below :**

| Column I | Column II |
|---|---|
| A Mesozoic Era | P Lower invertebrates |
| B Proterozoic Era | Q Fish & Amphibias |
| C Cenozoic Era | R Birds & Reptiles |
| D Paleozoic Era | S Mammals |

(1) A – Q, B – P, C – R, D – S

(2) A – R, B – P, C – Q, D – S

(3) A – P, B – Q, C – S, D – R

(4) A – R, B – P, C – S, D – Q

## 48. Match Column - I with Column - II.

| Column I | | Column II | |
|---|---|---|---|
| A | Adaptive radiation | P | Selection of resonance varities due to excessive use of herbicides and pesticides |
| B | Convergent evolution | Q | Bones offorelimbs in man and whale |
| C | Divergent evolution | R | Wings of butterfly and bird |
| D | Evolution by anthropogenic action | S | Darwin finches |

(1) A - S; B - R; C - Q; D – P

(2) A - R; B - Q; C - P; D - S

(3) A - Q; B - P; C - S; D – R

(4) A - P; B - S; C - R; D - Q

49. Match the columns I and II.

| Column I | Column II |
|---|---|
| A Oparin | P America |
| B J.B.S. Haldane | Q Russia |
| C Miller | R England |

(1) A – R, B – P, C – Q

(2) A – Q, B – R, C – P

(3) A – P, B – R, C – Q

(4) A – Q, B – P, C – R

# Title: Evolution

**50. Match the following columns.**

| Column I | Column II |
|---|---|
| A Wallace | P Archipelago |
| B Malthus | Q Influenced Darwin |
| C Hardy-Weinberg Law | R England |
| D Industrial melanism | S $p^2+2pq+q^2= 1$ |

(1) A – R, B – P, C – Q, D – S
(2) A – S, B – R, C – P, D – Q
(3) A – P, B – R, C – Q, D – S
(4) A – P, B – Q, C – S, D – R

# Title: Evolution

**51. Match the following columns.**

| Column I | | Column II | |
|---|---|---|---|
| A | Darwin | P | Branching descent |
| B | Lamarck | Q | Inheritance of aquired character |
| C | Pasteur | R | Disapproved spontaneous theory |
| D | De Vries | S | Mutational theory of inheritance |

(1) A – R, B – P, C – Q, D – S

(2) A – P, B – Q, C – R, D – S

(3) A – P, B – R, C – Q, D – S

(4) A – Q, B – P, C – R, D – S

**52. Match the columns I and II.**

| Column I | | Column II | |
|---|---|---|---|
| A | *Homo habilis* | P | 900 cc |
| B | *Homo neanderthalensis* | Q | 650-800 cc |
| C | *Homo erectus* | R | 1400 cc |
| D | *Homo sapiens* | S | 1350 cc |

(1) A – R, B – P, C – Q, D – S
(2) A – S, B – R, C – P, D – Q
(3) A – P, B – R, C – Q, D – S
(4) A – Q, B – R, C – P, D – S

# Title: Evolution

**53. Identify the correct match from column-I and column- II:**

| Column I | Column II |
|---|---|
| A. Origin of universe | P. 3 bya |
| B. Origin of Earth | Q. 20 bya |
| C. Origin of first cellular form of life | R. 4.5 bya |
| D. Origin of first noncellular form of life | S. 2 bya |

(1) A - P, B - Q, C - R, D – S

(2) A - Q, B - R, C - S, D - P

(3) A - Q, B - S, C - R, D – P

(4) A - P, B - R, C - Q, D - S

**54. Match the columns I and II.**

| Column I | | Column II | |
|---|---|---|---|
| A | Pasteur | P | Life only comes from life |
| B | Oparin | Q | Experimental verification of chemical evolution |
| C | S.L. Miller | R | Theory of chemical evolution |

(1) A – R, B – P, C – Q
(2) A – Q, B – R, C – P
(3) A – P, B – R, C – Q
(4) A – Q, B – P, C – R

**55. Match the following columns.**

| Column I | | Column II | |
|---|---|---|---|
| A | *Ichthyosaurus* | P | Caught in South Africa in 1938 |
| B | Coelacanth | Q | Fell to form coal deposits |
| C | Giant pteridophytes | R | Disappeared about 65 mya |
| D | Dinosaurs | S | Fish-like reptiles in 200 mya |

(1) A – R, B – P, C – Q, D – S
(2) A – S, B – R, C – P, D – Q
(3) A – P, B – R, C – Q, D – S
(4) A – S, B – P, C – Q, D – R

**56. Match the following**

| Column I | | Column II | |
|---|---|---|---|
| A | First to use fire | P | *Homo erectus* |
| B | Ape like primate | Q | *Dryopithecus* |
| C | Ancestor of modern apes | R | *Australopithecus* |
| D | Connecting link between ape and man | S | *Propliopethecus* |

(1) A – P, B – S, C – Q, D – R
(2) A – Q, B – R, C – P, D – S
(3) A – Q, B – S, C – R, D – P
(4) A – R, B – P, C – S, D – Q

# Title: Evolution

**57. Statement A : Wings of insects and bats are analogous.**

**Statement B : Wings of insects and birds are analogous.**

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

(3) Both statements are correct.

(4) Both Statements are incorrect.

**58. Read the statements A and statement B carefully to mark the correct options given below.**

**Statement A : All vertebrates develop a row of vestigial gill slits during embryonic stage.**

**Statement B : Embryos always pass through the adult stages of other animals.**

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

(3) Both statements are correct.

(4) Both Statements are incorrect.

# Title: Evolution



**59. Which of the following statements regarding the evolution of plants and animals are correct?**

**I. Reptiles evolved into birds**

**II. Fish with stout and strong fins could move on land and go back to water. This was about 350 million years ago.**

**III. Giant ferns fell to form coal deposits slowly.**

**IV. About 65 billion years ago dinosaurs died out.**

**V. Archaeopteryx is the connecting link between birds and reptiles.**

**The correct answer is**

(1) I and II (2) III and IV

(3) V and I (4) I, II, III, & V

# Title: Evolution

**60. Statement A : In natural selection, heritable variations enable better survival**

**Statement B : Australopithecus hunted with stone weapons**

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

(3) Both statements are correct.

(4) Both Statements are incorrect.

# Title: Evolution

**61. Statement A : Charles Darwin was on a sea voyage in a sail ship called H.M.S. Beagle**

**Statement B : Charles Darwin concluded that existing living forms share similarities to varying degrees not only among themselves but also with life forms that existed millions of years ago.**

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

(3) Both statements are correct.

(4) Both Statements are incorrect.

# Title: Evolution

**62. Statement A : Natural selection is an outcome of the differences in survival and reproduction among individuals that show variations in one or more traits.**

**Statement B : Adaptive forms of a given trait tend to become more common. Less adaptive ones beomes less common or disappear.**

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

(3) Both statements are correct.

(4) Both Statements are incorrect.

# Title: Evolution

63. Consider the following statements.

I. Homo habilis: The brain capacities were between 650-800cc.

II. Erect man or Homo erectus is skull was flatter than that of modern man.

III. Homo erectus probably ate meat

IV. Neanderthal man is regarded as most recent ancestor of today's man.

Choose the correct statements.

(1) I, III and IV
(2) I, II and III
(3) I and II
(4) I and III

# Title: Evolution

**64. Oparin of_____ and Haldane of_______ proposed that the first form of life could have come from preexisting non-living organic molecules.**

(1) Russia, Russia

(2) Russia, England

(3) England, England

(4) England, Russia

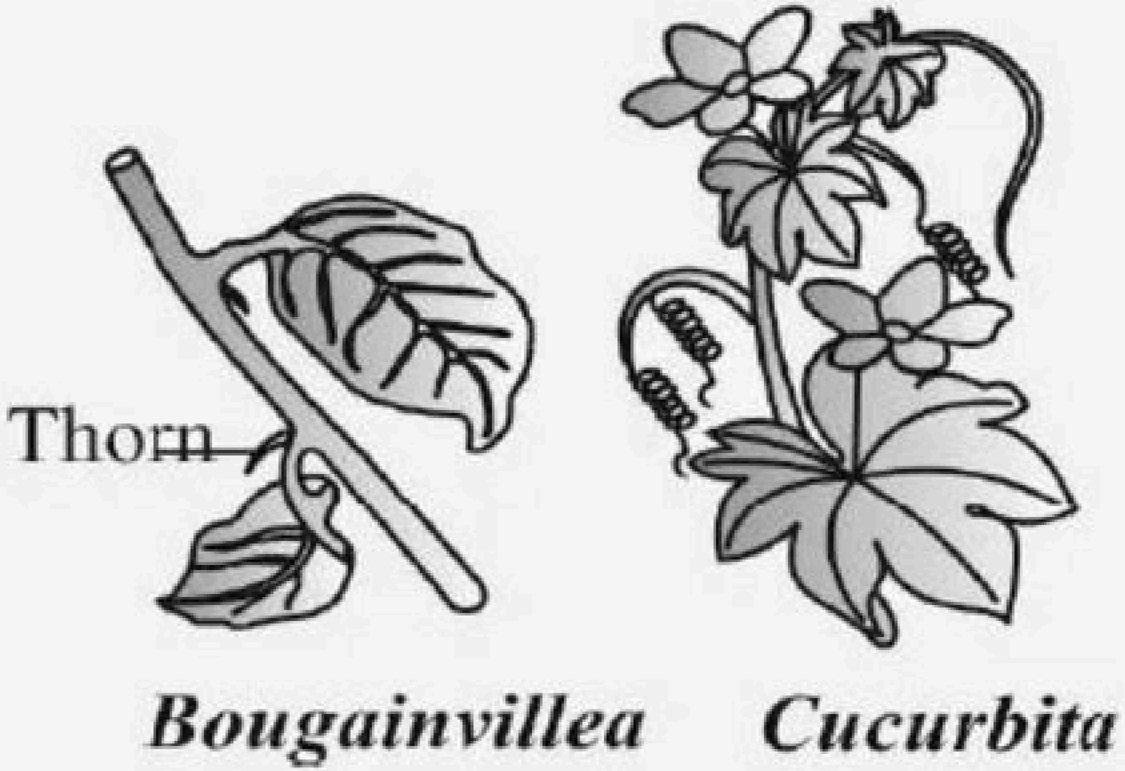
**65. The following diagram represents homologous organ, which part of the plant represents homology?**

(1) Root tendrils

(2) Leaf tendrils

(3) Stem tendrils

(4) Spine tendrils



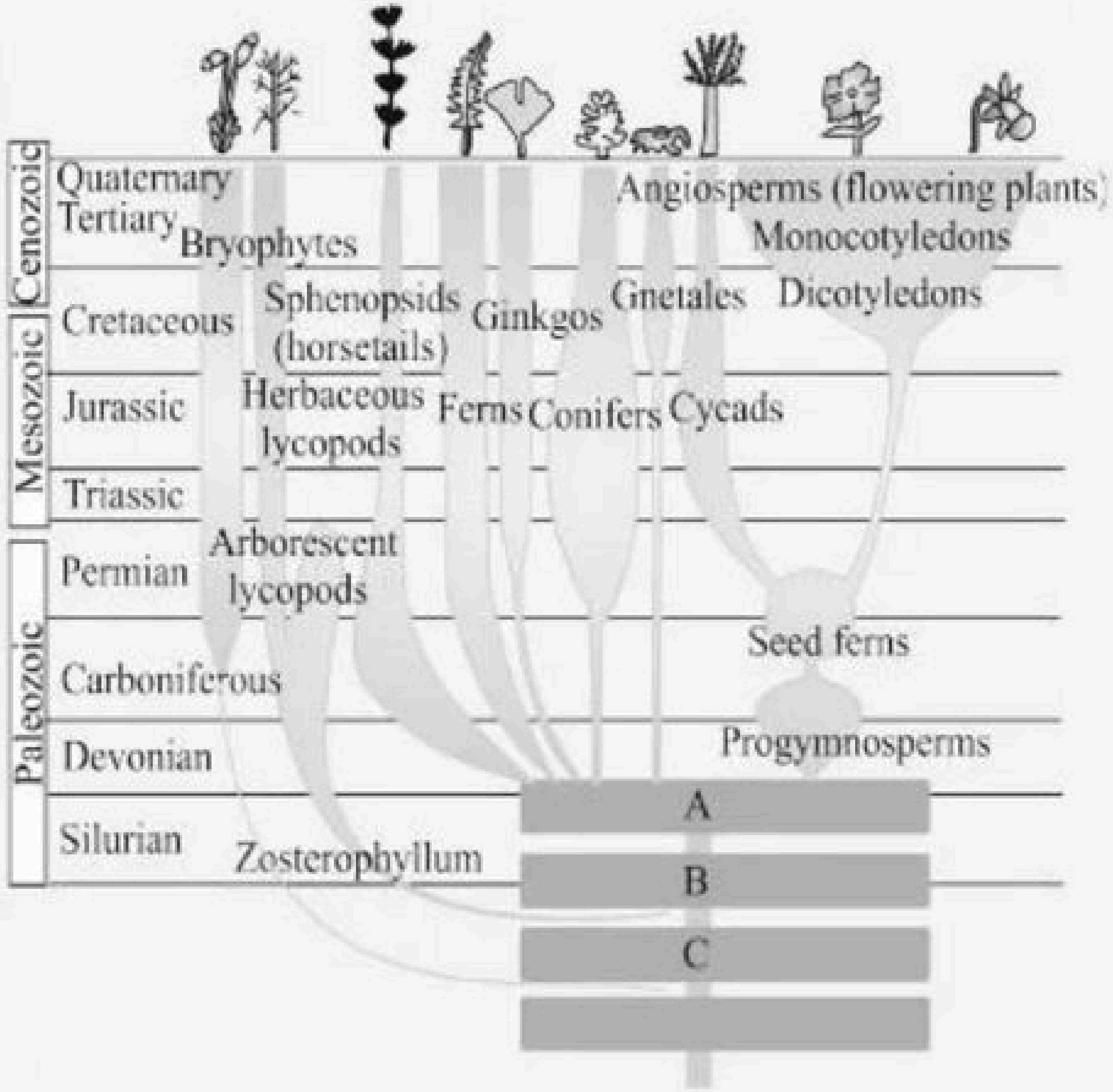
**66. Identify the missing ancestors.**

(1) A - Chlorophyte ancestor,B - Rhynia type,C - Tracheophyte

(2) A - Psilophyton,B - Rhynia type,C - Tracheophyte

(3) A - Psilophyton,B - Tracheophyte,C - Chlorophyte

(4) A - Psilophyton,B - Rhynia,C - Chlorophyte



# Title: Evolution

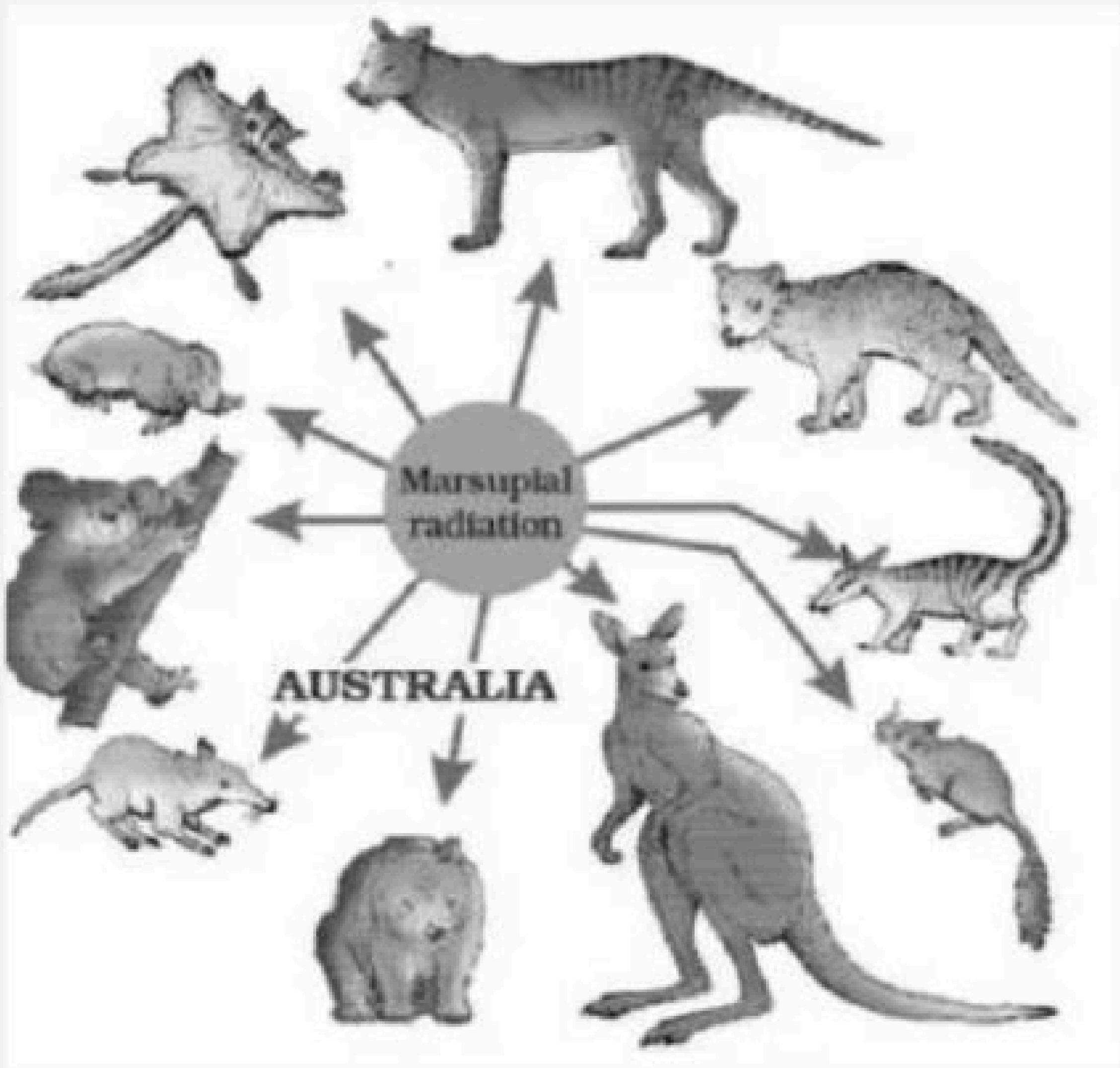
**67. What does the following diagram represent?**

(1) Natural selection

(2) Marsupial radiation

(3) Placental radiation

(4) Mutation



# Title: Evolution

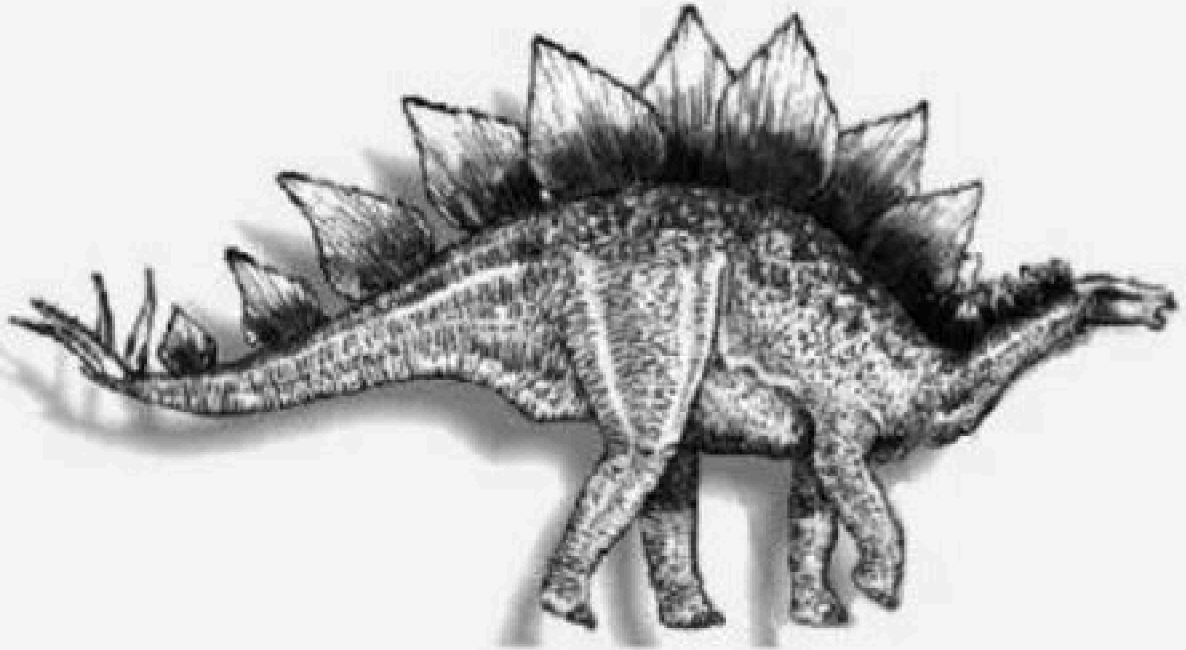
**68. Identify the following dinosaur.**

(1) Branchiosaurus

(2) Stegosaurus

(3) Archeopteryx

(4) Triceratops



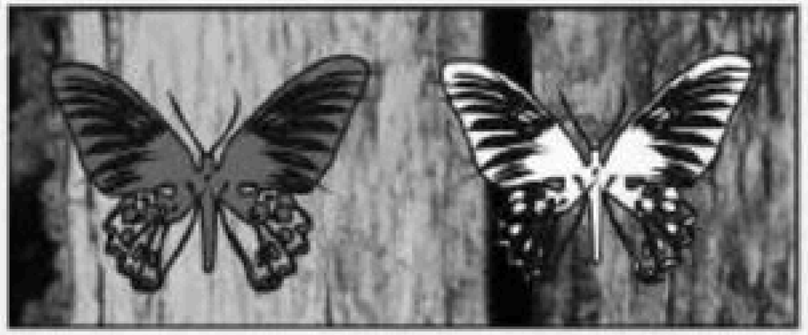
**69. The flowering insect represents a particular phenomenon in England. Identify the correct option.**

(1) Mutation

(2) Natural selection

(3) Adaptive radiation

(4) Convergent evolution



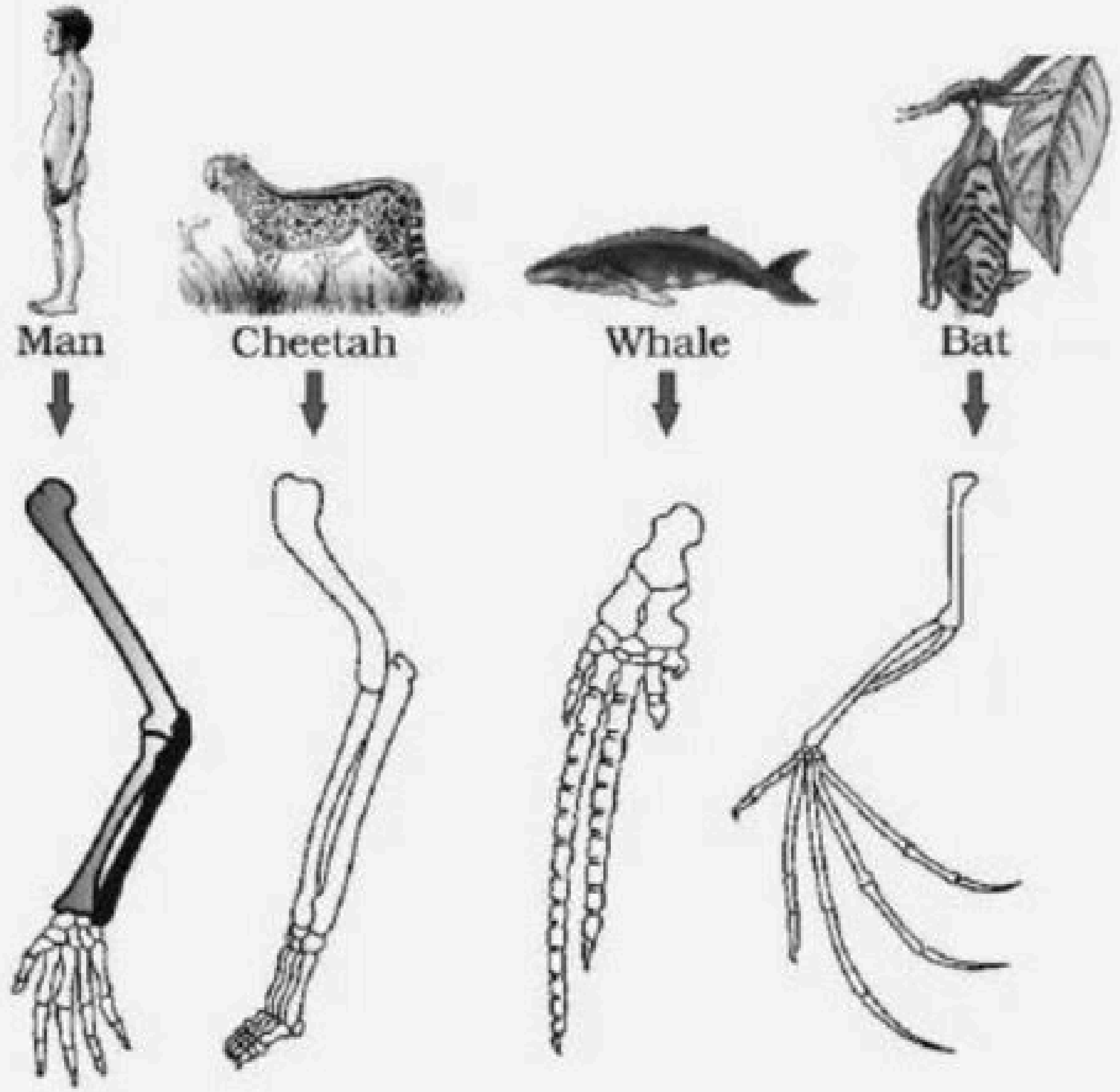
**70. The following diagram represents**

(1) Homology in plants

(2) Homology in animals

(3) Divergent evolution

(4) Both 2 & 3



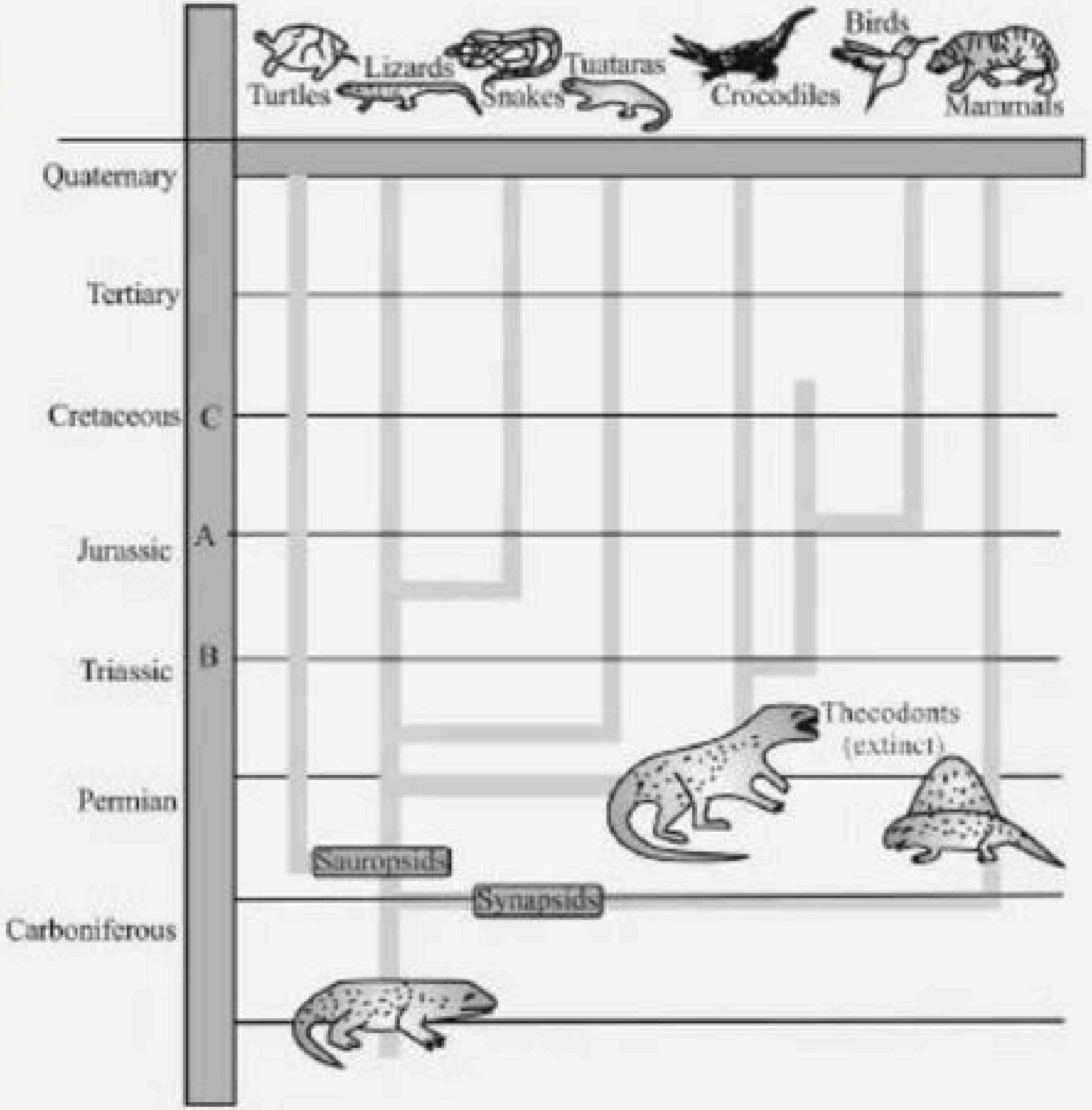
**71. Identify the missing time period.**

(1) A - 150, B - 100, C -200

(2) A - 150, B - 200, C -100

(3) A - 300, B - 350, C -250

(4) A - 250, B - 350, C -300



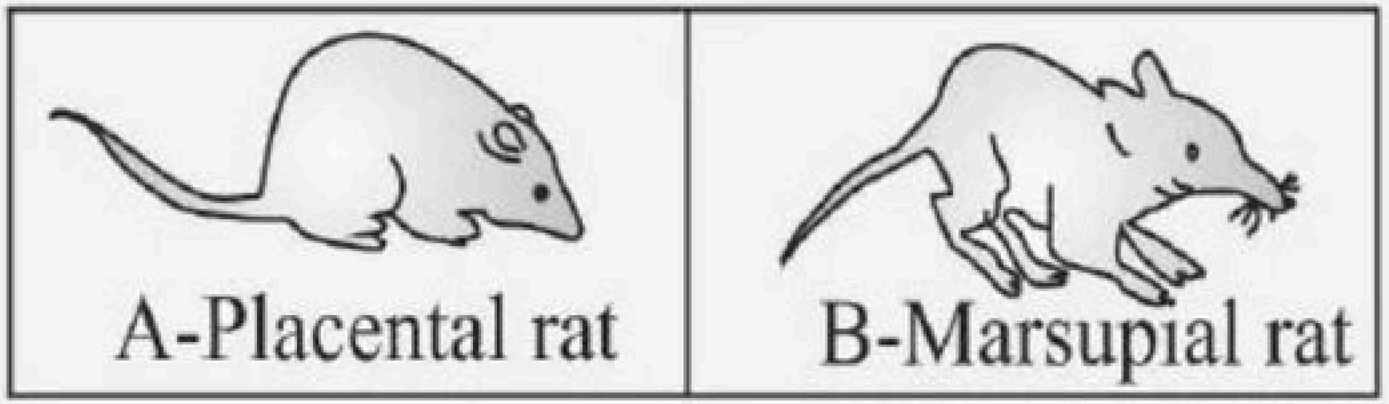
**72. The organism A and B are example of:**

(1) Convergent evolution

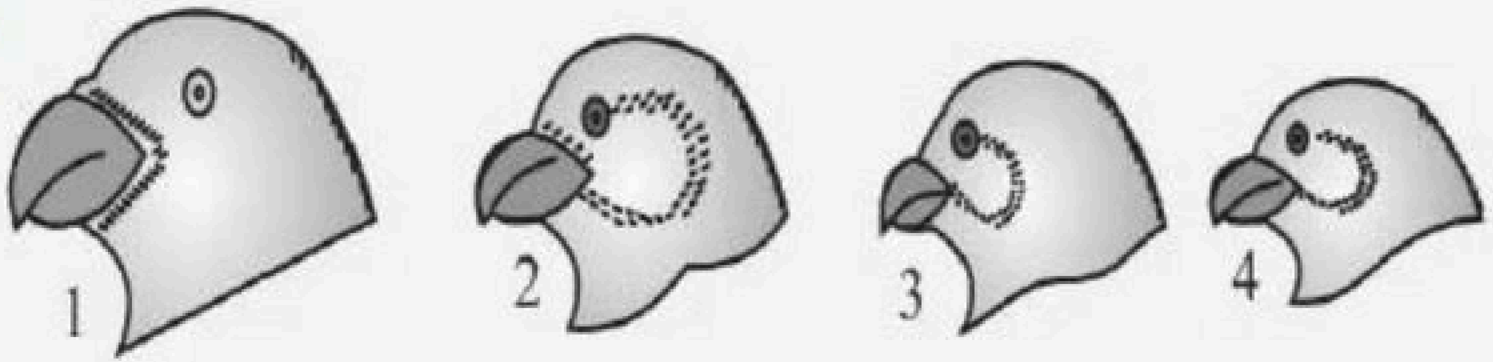
(2) Divergent evolution

(3) Adaptive convergence

(4) Both (1) and (3)



**73. What does the given image depicts?**




(1) Variety of beaks of Darwins finches

(2) Variety of birds of African island

(3) Variety of beaks of finches found in Galapagos island.

(4) Both (1) and (3)

# Title: Evolution

**74. The given figure shows an example of**

(1) Homologous organs

(2) Convergent evolution

(3) Analogous organs

(4) Vestigial organs

# Title: Evolution

**75. The given figure represents**

(1) Divergent evolution

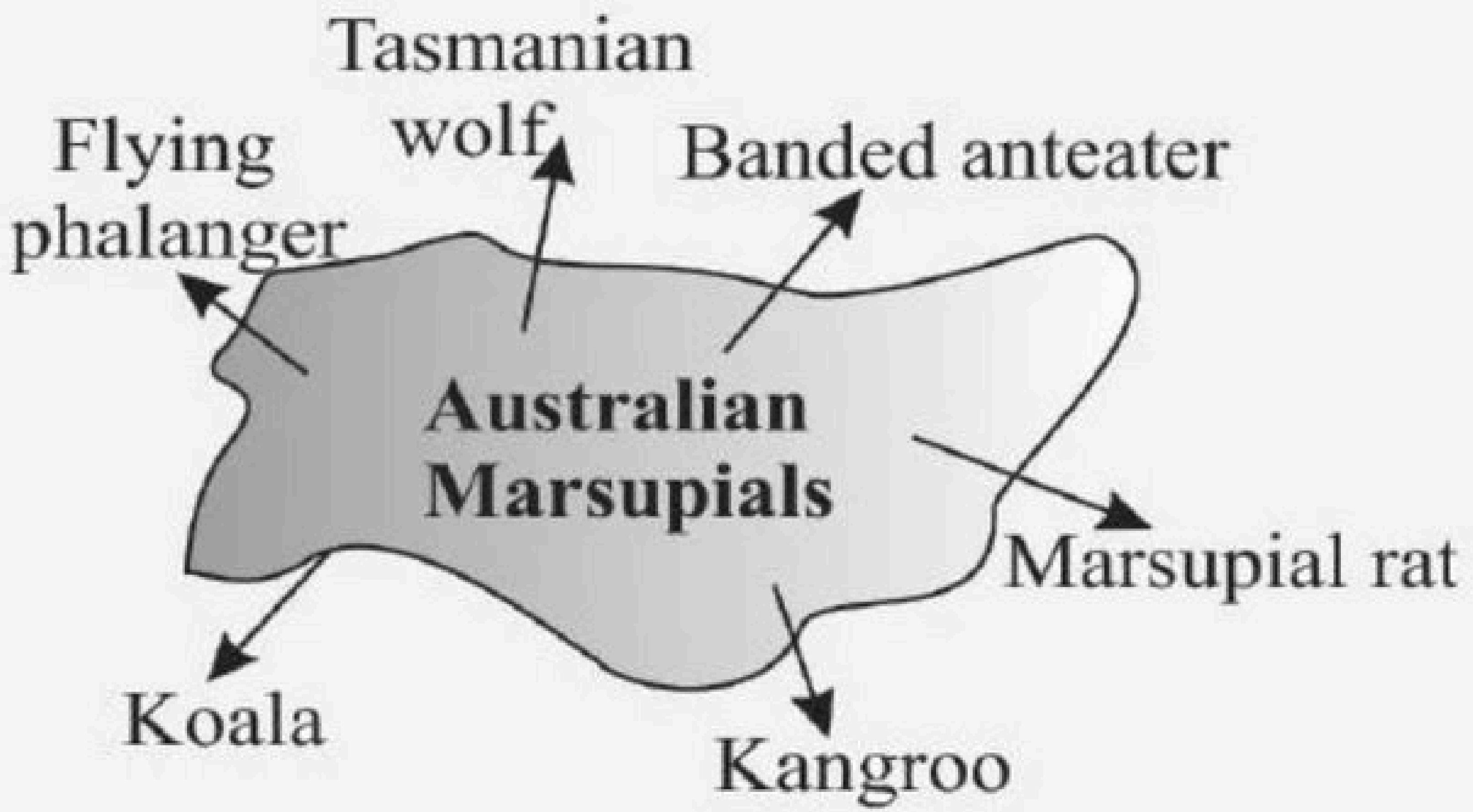
(2) Adaptive radiation

(3) Convergent evolution

(4) Classification

| Niche | Placental Mammals | Australian Marsupials |
|---|---|---|
| Burrower | Mole | Marsupial mole |
| Anteater | Anteater | Numbat(anteater) |
| Mouse | Mouse | Marsupial mouse |
| Climber | Lemur | Spotted cuscus |
| Glider | Flying squirrel | Flying phalanger |
| Cat | Bobcat | Tasmanian "tiger cat" |
| Wolf | Wolf | Tasmanian Wolf |

**76. The given diagram of marsupials of Australia provides an example of**



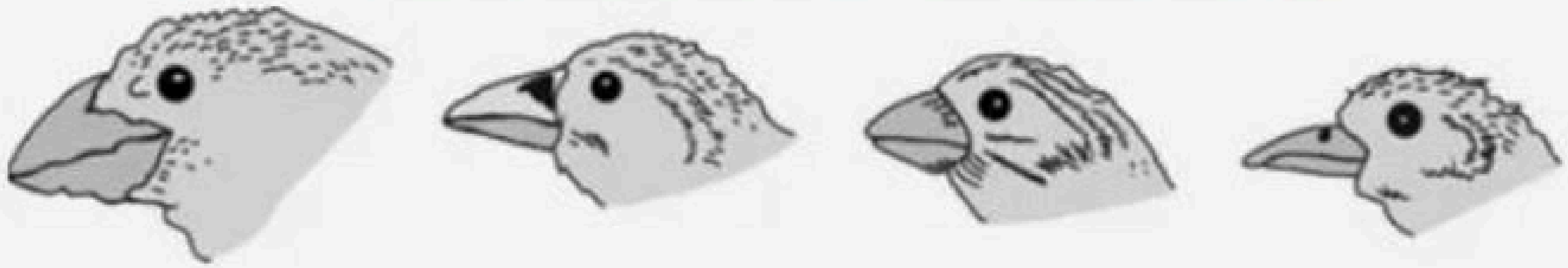

(1) Convergent evolution

(2) Parallel evolution

(3) Recapitulation

(4) Divergent evolution

**77. There is a difference in one morphological character in the birds. The part involved in the change is**



(1) Size of eye

(2) Shape of beak

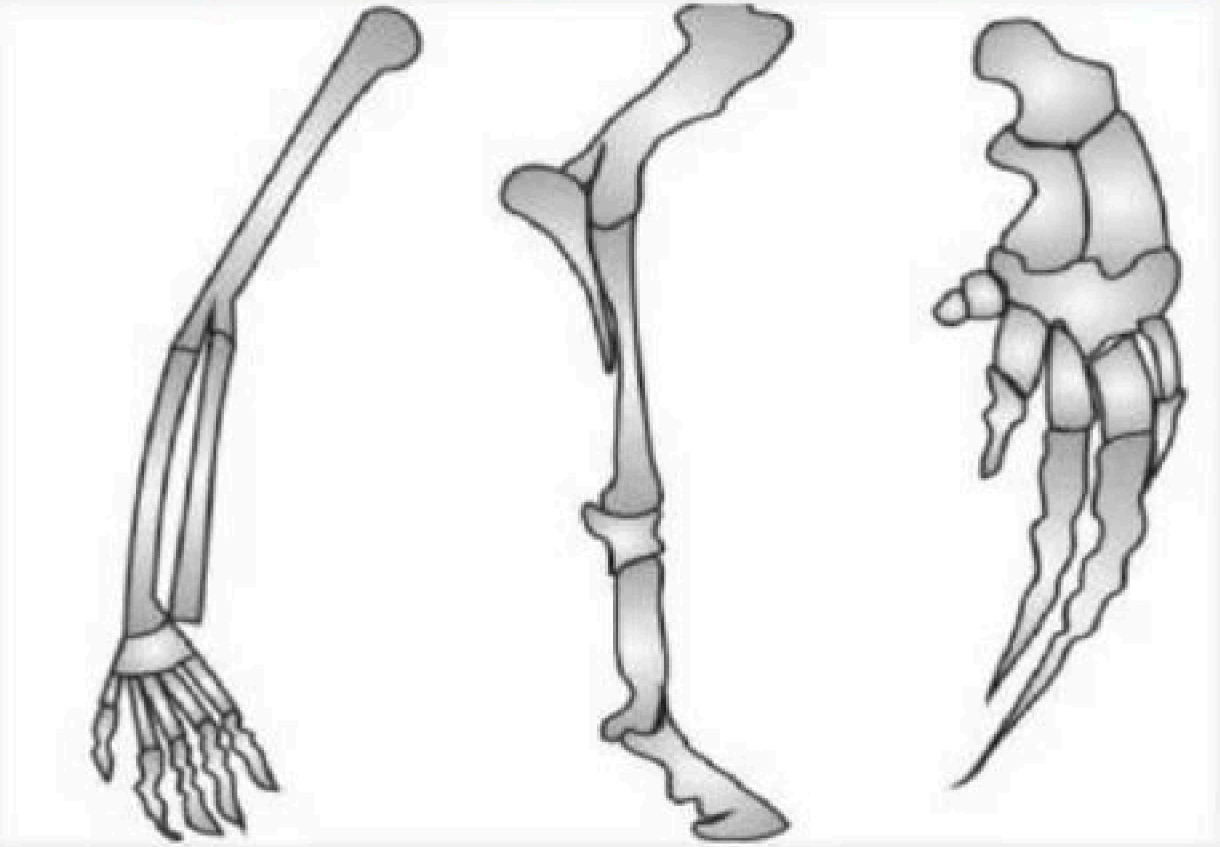
(3) Size of bird size/

(4) Size of wings.

# Title: Evolution

**78. The given bones in the forelimbs of three mammals figure shows.**

**For these mammals, the number, position, and shape of the bones most likely indicates that they may have**



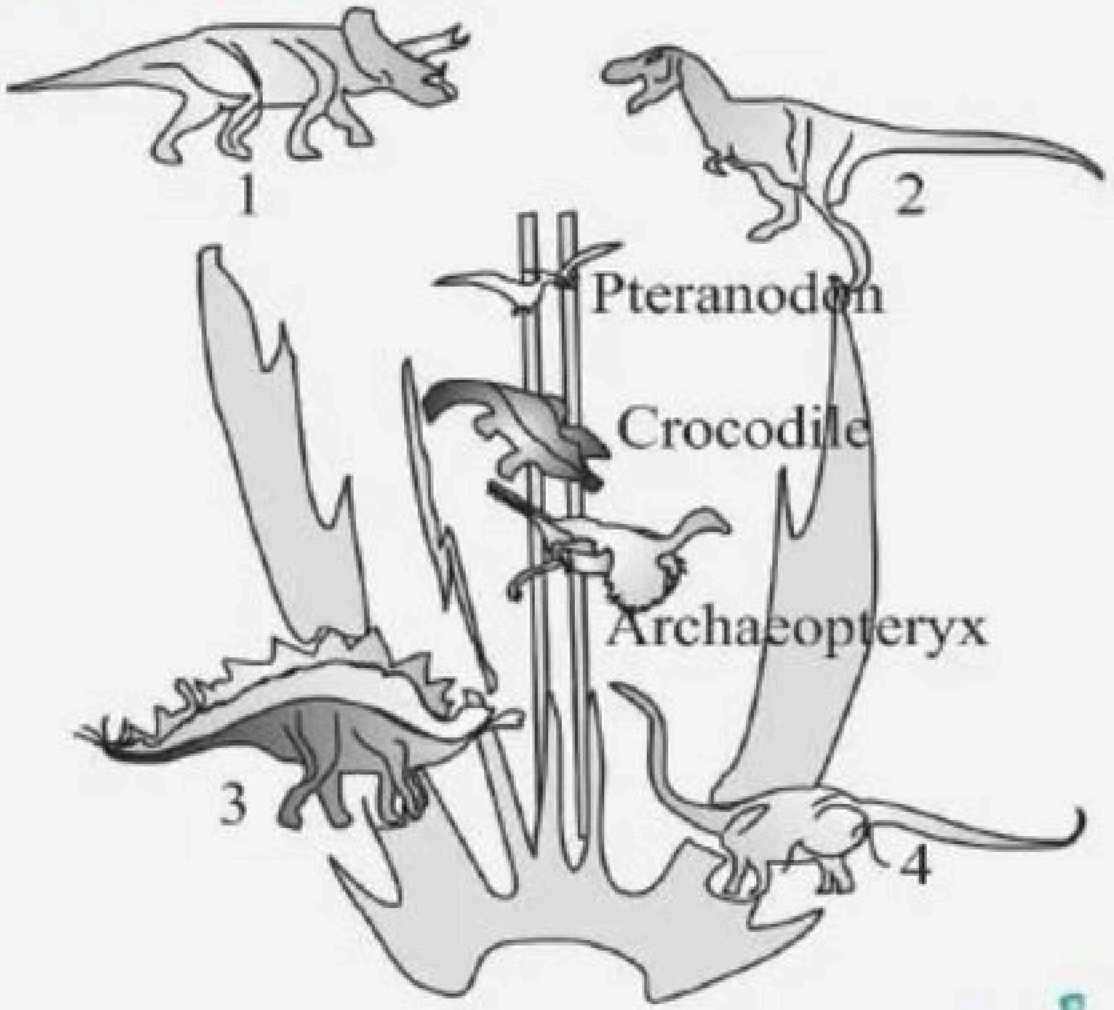
(1) Developed in a common environment.

(2) Developed from the same earlier species.

(3) Identical genotype

(4) Identical methods of obtaining food.

**79. A family tree of dinosaurs and their living modern day counterpart organisms like crocodiles and birds are given below. Identify 1,2, 3 and 4.**




(1) 1 - Stegosaurus,2 - Brachiosaurus, 3 - Tyrannosaurus, 4 - Triceratops

(2) 1 - Triceratops,2 - Tyrannosaurus, 3 - Stegosaurus, 4 - Brachiosaurus

(3) 1 - Stegosaurus, 2 - Triceratops, 3 - Brachiosaurus, 4 - Tyrannosaurus

(4) 1 - Triceratops, 2 - Brachiosaurus, 3 - Stegosaurus, 4 - Tyrannosaurus

# Title: Evolution

**80. Which of the following is the correct sequence of events in the origin of life.**

**i. Formation of protobionts**

**ii. Synthesis of organic monomers.**

**iii. Synthesis of organic polymers.**

**iv. Formation of DNA- based genetic systems.**

(1) i, ii, iii, iv

(2) i, iii, ii, iv

(3) ii, iii, i, iv

(4) ii, iii, iv, i

# Title: Evolution

**81. Forelimbs of cat, lizard used in walking; forelimbs of whale used in swimming and forelimbs of bats used in flying are an example of:**

(1) Analogous organs

(2) Adaptive radiation

(3) Homologous organs

(4) Convergent evolution

# Title: Evolution

**82. What was the most significant trend in the evolution of modern man (Homo sapiens) from his ancestors?**

(1) Binocular vision

(2) Upright posture

(3) Shortening of jaws

(4) Increasing cranial capacity

# Title: Evolution

**83. The flippers of the Penguins and Dolphins are the example of the**

(1) Adaptive radiation

(2) Natural selection

(3) Convergent evolution

(4) Divergent evolution

**84. Study the following table and identify the correct option for A, B and C.**

| | Fossil | Cranial capacity | Time of appearance |
|---|---|---|---|
| I | Australopi-thecus | A | 5 mya |
| II | Homo sapiens neandertha-lensis | 1400 cc | B |
| III | C | 900 cc | 1.5 mya |

(1) A - 700cc, B - 3400 years ago, C - Homo erectus

(2) A - 500cc, B - 11000 years ago, C - Homo habilis

(3) A - 800cc, B - 50,000 years ago, C - Ramapithecus

(4) A - 500cc, B - 100,000 years ago, C - Homo erectus

# Title: Evolution

**85. The ancestors of modern day frogs and Salamanders are**

(1) Jawless fish

(2) Coelocanth

(3) Icthyophis

(4) Amphioxus

# Title: Evolution

**86. Hardy - Weinberg principle is expresed by following equation:**

(1) $p^2 - q^2$

(2) $p^2 + q^2$

(3) $p^2+2pq+q^2$

(4) $p^2 - 2pq - q^2$

**87. Select the correct order of geological time scale of earth.**

(1) Palaeozoic ⟶ Archaeozoic ⟶ Cenozoic

(2) Archaeozoic ⟶ Palaeozoic ⟶ Proterozoic

(3) Palaeozoic ⟶ Mesozoic ⟶ Cenozoic

(4) Mesozoic ⟶ Archaeozoic ⟶ Proterozoic

# Title: Evolution

**88. The first non-cellular form of life could have originated ____ billion years back.**

(1) 4.5

(2) 2

(3) 3

(4) 4

# Title: Evolution

**89. The correct sequence of evolution of man:**

**I. Homo habilis**

**II. Java Man**

**III. Modern Man**

**IV. Dryopithecus**

(1) II, I, III, IV

(2) IV, I, II, III

(3) I, II, III, IV

(4) I, II, IV, III

# Title: Evolution

**90. In a population of 1000 individuals 360 belong to genotype AA, 480 to and the remaining 160 to aa. Based on this data, the frequency of allele A in the population is:**

(1) 0.4

(2) 0.5

(3) 0.6

(4) 0.7

**91. In a species, the weight of newborn ranges from 2 to 5 Kg 97%. of the newborn with an average weight between 3 to 3.3 kg survive whereas 99% of the infants born with weight from 2 to 2.5 kg or 4.5 to 5kg die.**

**Which type of selection process is taking place?**

(1) Directional Selection

(2) Stabilizing Selection

(3) Disruptive Selection

(4) Cyclical Selection

# Title: Evolution

**92. The frequency of two alleles in a gene pool is 0.19 (1) and 0.81(1). Assume that the population is in Hardy-Weinberg equilibrium. The percentage of heterozygous individuals in the population is**

(1) 0.81

(2) 0.31

(3) 0.19

(4) 1

# Title: Evolution

**93. About how many years ago Drypithecus and Ramapithecus were existed?**

(1) 15 million years ago

(2) 15 billion years ago

(3) 15 thousand years ago

(4) 18,000 years ago

**94. The Hardy-Weinberg law of equilibrium was based on the following**

(1) Random mating, natural selection, gene flow

(2) Random mating, genetic drift, gene flow

(3) Non-random mating, mutation, gene flow

(4) Random mating, no mutation, no gene flow

# Title: Evolution

**95. About_____ the first cellular forms of life appeared on earth.**

(1) 3000 (mya)

(2) 2000 (mya)

(3) 2000 (bya)

(4) 3000 (bya)

# Title: Evolution

**96. All organisms share the same genetic code. This statement is in evidence that**

(1) All the organisms looks similar

(2) The evolution which has occured was very fast

(3) The evolution occurs gradually

(4) All the organisms are descended from a common ancestor

**97. From the original seed-eating features, many other forms with altered beaks arose, enabling them to become**

(1) Insectivorous

(2) Vegetarian finches

(3) Tortoise

(4) Both 1 and 2

# Title: Evolution



**98. Randomly mating population has an established frequency of for organisms that are homozygous recessive for a given trait. The frequency of this recessive allele in the gene pool is**

(1) 6

(2) 36

(3) 0.6

(4) 16

# Title: Evolution

**99. Which is incorrectly matched?**

(1) Jawless fish probably evolved around 350 mya.

(2) 320 mya - Jawless fishes evolved

(3) 2000 mya - First cellular form of life appeared

(4) 65 mya - Dinosaurs disappeared

# Title: Evolution



**100. The most accepted line of descent in human evolution is**

(1) Australopithecus → Ramapithecus → Homo sapiens → Homo habilis

(2) Homo erectus → Homo habilis → Homo sapiens

(3) Ramapithecus → Homohabilis → Homo erectus → Homo sapiens

(4) Australopithecus → Ramapithecus → Homo erectus → Homo habilis → Homo sapiens

# THANK YOU

BIOLOGYLIVE